

Digital Companion for Field Drug Testing

Smart India Hackathon 2026 | Software | MedTech / BioTech / HealthTech

A clearer field record.

A traceable path to laboratory confirmation.

FieldSeal - submission prototype

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Problem statement organisation: Ministry of Home Affairs (NCB)

Prototype scope and intended capabilities are separated throughout. No agency endorsement or validated identification performance is claimed.

The problem and our response

Support the existing field workflow without changing the kit or overstating the result.

Record the observation

Field colour readings can be affected by illumination, camera processing, kit condition and interpretation. Capture the image, kit lot, operator and case reference together.

Preserve a reviewable trail

Link capture, evidence bag, transfers and laboratory follow-up under one record ID. Use hashes and signatures to make changes detectable within a stated trust model.

Abstain when evidence is weak

An unsuitable capture or unvalidated input remains inconclusive, with a reason and a retake path. A phone image alone does not establish substance identity.

Keep confirmation independent

Presumptive field results require appropriate laboratory confirmation. GC-MS is one possible analytical method, not the only universally required method.

A complete prototype journey

Demonstration scenarios are labelled. Uploaded photographs are recorded without invented classifications.

- | | |
|--------------------|---|
| 1 Register | Enter case, bag and operator details; select a kit and block expired lots. |
| 2 Capture | Upload a photo or select a synthetic sample; show reference, lighting and alignment checks. |
| 3 Review | Show a demonstration indication, no indication or Inconclusive result with a reason. |
| 4 Seal | Persist an immutable record payload; calculate SHA-256 and sign with a demo server key. |
| 5 Verify | Export a signed bundle; detect modified payloads or custody entries during verification. |
| 6 Follow up | Record custody transfers and link a manually entered laboratory outcome. |

Architecture with a clear trust boundary

Prototype implementation target - independently testable frontend, API and record integrity.

Web interface

React + TypeScript
Responsive capture, records, verifier and kit views.
Local drafts and a visible offline queue; synchronization when connected.

Integrity foundation

SHA-256 content digests
ECDSA P-256 demo server signatures
Chained custody events and idempotent record creation.
Private keys stay out of Git.

API and persistence

FastAPI + validated request models
SQLite for a reproducible local demo.
PostgreSQL migration path for a future multi-user deployment.

Production work remains

Authentication and permissions, encrypted storage and key management, native hardware attestation, trusted timestamps, privacy review and agency integration.

Honest claims make a stronger submission

Corrections applied to the original deck.

Tamper-proof / court-ready

Tamper-evident under a defined key and record-storage trust model. Admissibility is not guaranteed by software.

Any phone becomes the same instrument

Calibration is an engineering goal. Validate device, lighting, reagent and card variation before claiming equivalence.

Validated on 5,000 images / 80% savings

Remove as achieved claims. No supporting dataset or measured timing study was supplied.

100% offline and hardware-backed signing

Separate intended native production features from browser draft storage and demo server signing.

Public QR exposes each case

Use privacy-preserving verification in production; avoid disclosing case details through public links.

Validation before operational use

The prototype demonstrates workflow and integrity. Predictive performance must be measured separately.

Build a labelled dataset

Obtain lawful, authorized kit-specific reference captures with independent laboratory outcomes. Record device, lighting, kit lot, expiry and reference-card details.

Keep evaluation independent

Separate training, validation and final test data by sample/session and kit/device where appropriate. Report per-class metrics and uncertainty; avoid demo-only accuracy claims.

Measure the failure cases

Test false indications, missed indications, ambiguous readings and abstentions. Include interference, poor illumination, missing references and unsupported devices.

Test records as well as models

Check altered payloads, deleted/reordered events, duplicate sync requests, expired kits and missing captures. Review access, retention and incident handling before a pilot.

Evidence, references and next milestones

Source material corrected for relevance and claim strength.

References

SWGDRUG - Approved Recommendations, Edition 8.2 (27 June 2024). Colour testing has limited discriminating power; identification uses an appropriate analytical scheme.

swgdrug.org/approved.htm

Bharatiya Sakshya Adhiniyam, 2023 - Section 63 and applicable certificate requirements. A generated bundle is not a completed statutory certificate; implementation requires qualified legal review.

indiacode.nic.in - official legislation

Next: finish and test the prototype; publish reviewed code and demo links; establish an authorized dataset and validation protocol before deploying a classifier.

Unrelated satellite/LoRA references, unsubstantiated market figures and the duplicate draft page have been removed. Submission links remain pending until verified.